

Aerogel

Aerogels are a class of synthetic porous materials derived from a gel, in which the liquid component has been replaced with a gas without collapsing the gel structure by freeze-drying. It can be made from silica, carbon, iron oxide, gold, copper, polymer, etc.

The final product is extremely porous (80-98% porosity) with very little solid material; up to 99.8% of the aerogel may have nothing but air. It has a typical density of about 0.001 gm cm^{-3} . Its thermal conductivity is extremely low, about 0.017 W/mK , which makes the material an ideal insulator. It can be made transparent.

By reducing heat losses and simultaneously being transparent, aerogels allow a solar plant to operate at higher temperature and at higher efficiency without using any vacuum device. These advantages make an aerogel assisted solar thermal plant very economical and eliminates lot of maintenance problems.

Concentrated high efficiency solar TE generator

A solar thermoelectric generator (STEG) is a solid-state device that can convert solar energy at around 15% efficiency. It has three sections: solar absorber, thermoelectric generator, and the thermal management system comprising insulation, heat exchangers, and vacuum/aerogel enclosure, etc. There are no moving parts and there is no need of high-temperature operating fluids. Its robustness in harsh temperatures makes it very useful for standalone power conversion or making hybrid solar thermal power generator in conjunction with a PV system.

The efficiency of STEG depends on both the efficiency of solar absorber and on the thermoelectric efficiency of the device. There are mainly two approaches to increase the efficiency of STEG devices: increasing temperature difference

between the hot and cold ends and using improved materials with ZT more than 1. Recently, several nanostructured materials have been developed that have higher ZT val-

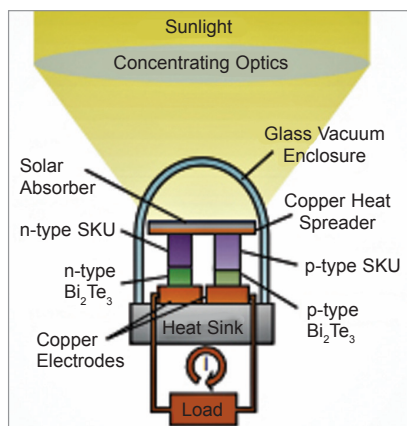


Fig. 11: Schematic arrangement of concentrated STEG system

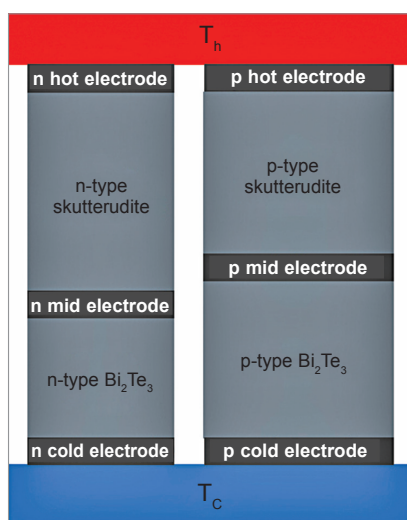


Fig. 12: Arrangement of composite segmented thermoelectric LEG

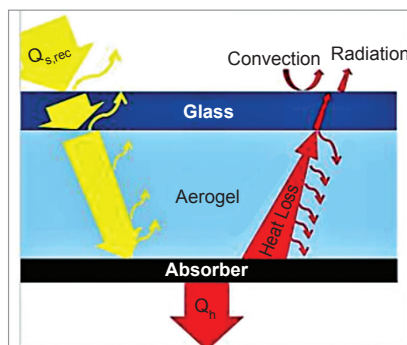


Fig. 13: Transparent aerogel for insulating STEG

ues suitable for STEG.

There are two routes to increase the temperature difference in a STEG: first, optical concentration of sunlight enabling to increase the heat flux at the absorber surface, and second by providing thermal concentration where the area of a highly thermally conducting absorber is greater than area of the thermoelectric legs increasing the heat flux through the legs.

It is possible to construct a durable STEG device with more than 15% efficiency using improved material consisting of a segmented n-type leg composed of skutterudite and La_1Te_4 and p-type leg of skutterudite and $\text{Yb}_{14}\text{MnSb}_{11}$, and also using nano-structured Bi_2Sb_3 based alloys (all these three types have an effective ZT of about 1.4 to 1.6 at 800K).

Operating temperatures across $T_{\text{hot}} - T_{\text{cold}} = 900^\circ\text{C} - 200^\circ\text{C}$ are maintained using fresnel lens to concentrate sunlight. In order to reduce heat losses from the solar collector a vacuum system is provided, which is costly. Skutterudite is a type of arsenide mineral having general formula as TPn_3 , where T is a transition metal like Co, Rh, or Ir, and Pn is Sb, As, or P. Typical arrangement of STEG system is shown in Fig. 11 and Fig. 12.

Another excellent way to reduce heat losses is by using high-temperature transparent aerogels for insulation instead of vacuum system, as shown in Fig. 13.

Aerogel, being transparent but an excellent insulator, allows sunlight to enter but blocks the heat from escaping from the receiver of the STEG. It offers many advantages like higher efficiency, minimises heat loss, boosts solar thermal conversion, eliminates costly vacuum system and its maintenance cost. Due to efficient thermal transport system provided by aerogels, most of the solar radiation is absorbed by the cermet composite pad attached to the thermoelectric elements.

It is desirable to thermally insu-